unit-5

modified

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Title :- Finding shortest Bus Route and minimum Travel time Between source and Destination.

modification from the original project. The original project calculates the minimum Travel time from one source stop To all reachable stops. The modified project accept a specific destination and also displays The Complete sequence of stops on The shortest route.

problem statement
Represent bus stop are vertices and bus routes are Weighted edges. where weight represents Non-Negative Travel times. Give a source stop and a destination stop. Use Dijkstra's algorithm To calculate The minimum Travel time and display The complete shortest path. if The destination, cannot be reachead. display a suitable message

solution
use an adjancy list and a min-prority queue as in The original version. in addition To The distance array. maintain a parent array. whenever a shorter distance To a stop is found. update its distance and store The proceding store in array.

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Algorithm

1) start

2. Read The Number of stops and route Then buid the weigted adjency list.

3. Read and validate The source and destination stops.

4. initialize all distance To infinity. all parent value to -1. and set The source distance to zero.

5. insert The source into a min-prority queue.

6. while queue is not empty. remove The stop with the smallest distance and skip outdated entries. stop early if the destination is reached with its current shortest distance.

7. for each neighbor. if The shorter distance is found. update its distance .set its parent To The current setop and insert the updated entry into The queue.

8. if The destination distance is infinity report That No route exists.

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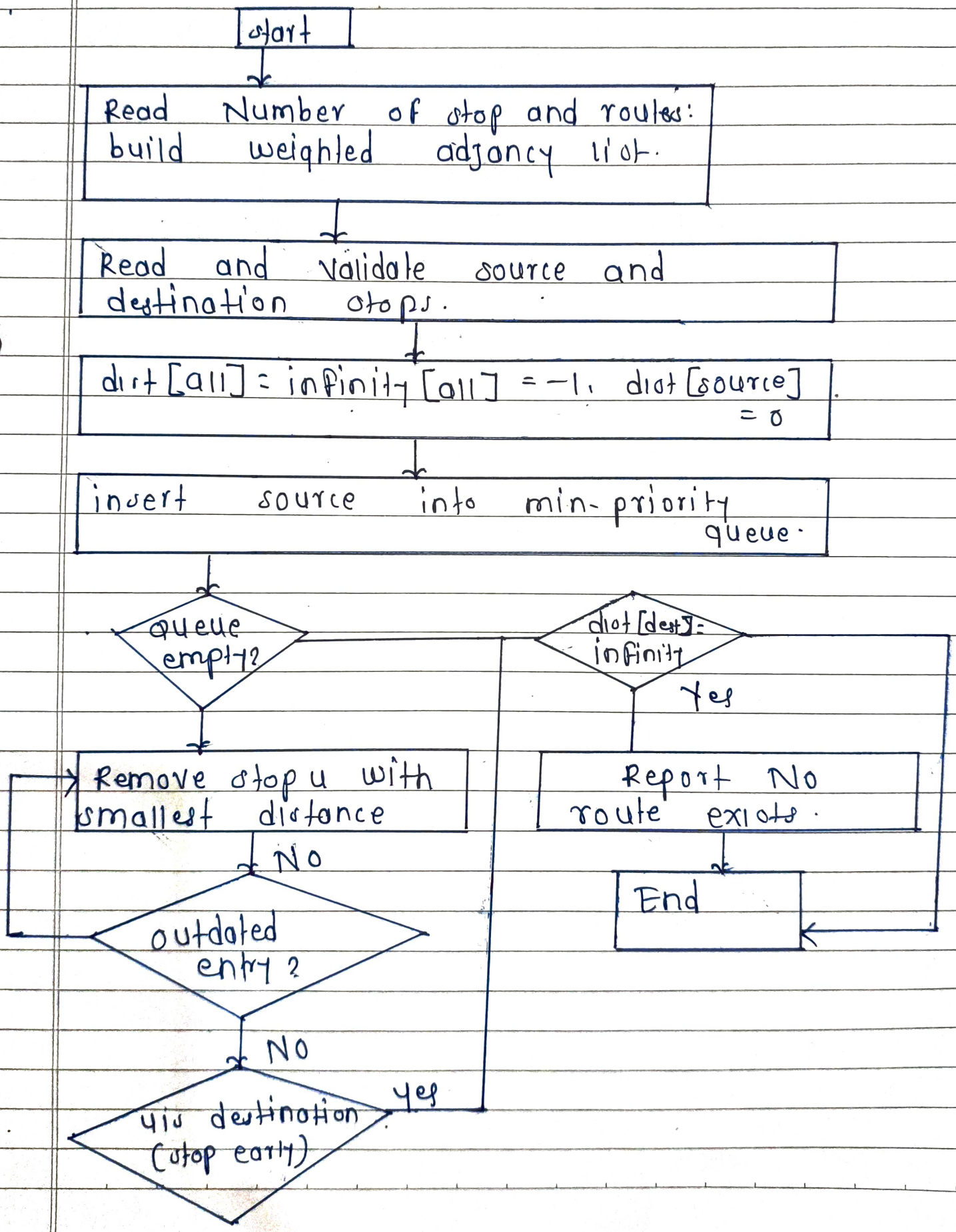
9. otherwise . Trace parent links backward from the destimation to build the path.

10. Reverse The path and display it with The minimum Travel time.

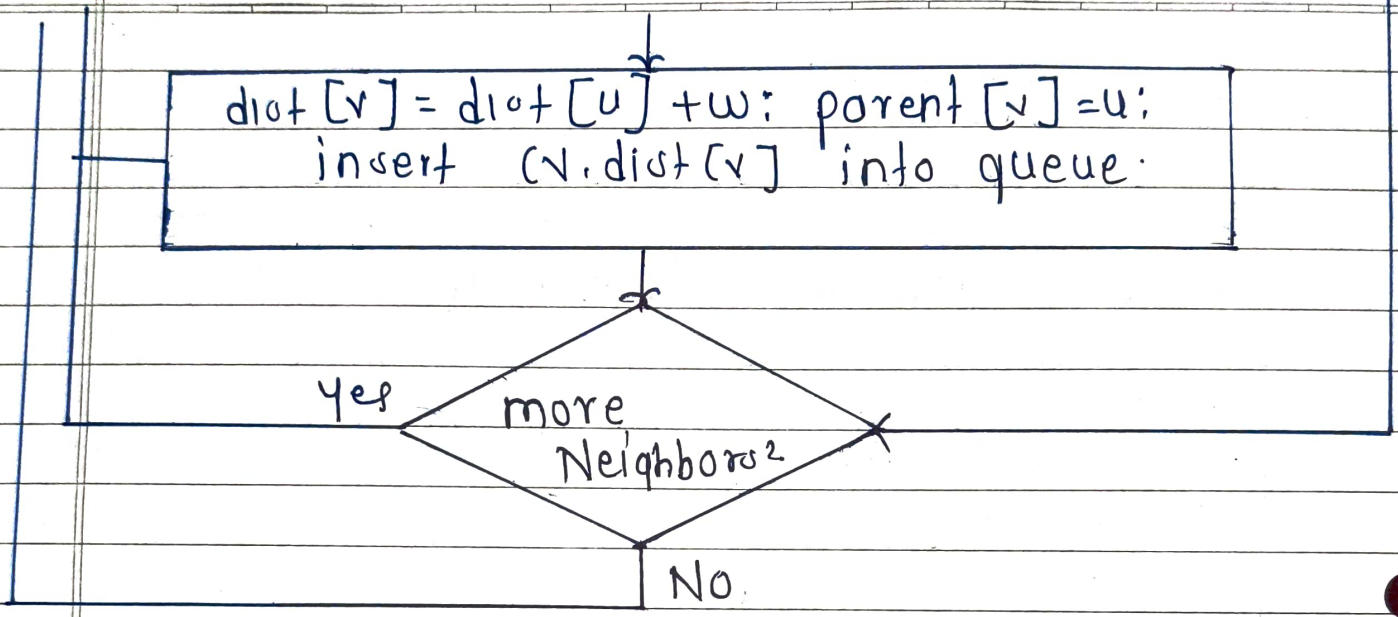
11. stop.

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Flowchart



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dist[v] = dist[u] + w; parent[v] = u;
insert (v, dist[v]) into queue.
more Neighbors?
Yes
No